Testing Measurement Invariance with Continuous Covariates: Comparing Moderated Nonlinear Factor Analysis and Structural Equation Model Trees

Leonie Hagitte[1,2,3] and Andreas M. Brandmaier[2,3,4]

[1]Goethe University Frankfurt

[2]Max Planck Institute for Human Development, Berlin, Germany

[3]MSB Medical School Berlin, Berlin, Germany

[4]Max Planck UCL Centre for Computational Psychiatry and Ageing Research, Berlin, Germany

Abstract

Establishing measurement invariance (MI) is essential for valid psychological assessments, yet measurement parameters may vary continuously across observed characteristics, resulting in measurement noninvariance (MNI) (Bauer et al., 2020). Moderated nonlinear factor analysis (MNLFA) models such variation but requires prespecifying its functional form (Bauer, 2017; Kolbe et al., 2024). Structural equation modeling (SEM) trees offer a nonparametric alternative using recursive partitioning (Arnold et al., 2021; Brandmaier et al., 2013, 2016). A Monte Carlo simulation compares linear MNLFA, quadratic MNLFA, and SEM trees using single-factor confirmatory factor analysis models with loading and intercept moderation under linear, quadratic, sigmoid, and null transformations. Population models, moderation magnitudes, and sample sizes are varied. SEM trees receive only the true moderator; a sensitivity analysis introduces ten irrelevant predictors. An empirical application uses CASP-12 data from SHARE Wave 9 (Börsch-Supan et al., 2013; Knesebeck et al., 2005; SHARE-ERIC, 2024; Wiggins et al.,2008). MNLFA detection performance depends on functional-form specification, whereas SEM trees show consistent sensitivity across moderation functions. Under scalar-only MNI, both MNLFA and SEM trees exhibit inflated metric MI rejection rates despite invariant loadings, with stronger inflation for SEM trees. In the empirical application, both approaches identify age and cultural background as sources of CASP-12 measurement heterogeneity. SEM trees provide flexibility when moderation functions are unknown but exhibit limitations in distinguishing metric from scalar MNI. Although generally less pronounced, similar classification problems occur with MNLFA. These findings highlight the complementary strengths and limitations of SEM trees and MNLFA in assessing continuous MNI.

Keywords: Measurement invariance, moderated nonlinear factor analysis, structural equation model trees, continuous moderators, Monte Carlo simulation

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Comparing persons or groups on psychological constructs, like intelligence, well-being, attitudes or personality, is one of the most basic acts in the social and behavioral sciences, yet it rests on an assumption that is all to often not made explicit: that the measure itself means the same thing for everyone being compared. Unlike weight or height of a person, latent constructs like these cannot be observed directly. Instead, they are inferred from indicators, typically responses to self-report items, using measurement models. If these measures differently work for the people or contexts being compared, one could say, if the same construct is measured with a different yardstick for different individuals, then any observed difference becomes ambiguous and ultimately unusable for solid inferences. It may reflect a genuine difference in the construct, an artifact of how it was measured, or some blend of both, and there is no way to disentangle the two after the fact (Flake and Fried 2020; Flake et al. 2017; Cheung and Rensvold 2002; Meredith 1993). This is the problem of measurement invariance (MI). The question of whether a measurement model operates equivalently across individuals, groups, contexts, or occasions (Meredith 1993). Far from just a technical requirement, MI is essential for comparing latent means, variances, covariances, or structural relations such as regressions on a latent level; which is to say, a precondition for much correlational, experimental, and longitudinal research designs (Putnick and Bornstein 2016; Little 1997).

Yet the standard approach for testing MI assumes something that is often untrue, namely that the source of measurement heterogeneity is categorical. Conventional approaches represent heterogeneity through predefined groups, e.g., sex, treatment arm, or age group, across which measurement parameters are constrained and compared. This works well when group boundaries are theoretically meaningful, but many of the most important candidate sources of measurement heterogeneity, such as age, are continuous, and their

relationship to measurement parameters is rarely known in advance. <!– Forcing a continuous variable into discrete categories requires often arbitrary cutoffs and can obscure exactly where and how measurement breaks down; assuming a particular functional form instead risks the opposite error, misrepresenting heterogeneity that does not follow that form. – > this doubles nearly exactly with a part later –> This leaves researchers with a genuine methodological bind: how can measurement heterogeneity across a continuous covariate be detected and characterized without committing in advance to either discrete groups or a specific functional shape? Two recent approaches, moderated nonlinear factor analysis (MNLFA) and Structural Equation Model trees (SEM trees), offer different answers to this question, and it is their comparative strengths and limitations that this paper investigates.

Classical Measurement Invariance Testing

Structural equation models (SEMs) are a principled framework for defining latent constructs and their relations to observed indicators (Kline 2023; Bollen and Noble 2011; Tarka 2018). Within a SEM, the measurement model specifies how latent factors relate to their manifest indicators, separating construct-related variance from measurement error and thereby providing a basis for evaluating measurement quality (Bollen 1989).

Classical MI testing builds on this framework in a confirmatory way. Researchers specify a measurement model, typically a confirmatory factor analysis (CFA), reflecting substantive hypotheses about the construct, then partition the sample by a categorical grouping variable and test increasingly restrictive equality constraints across groups. This approach, multiple-group CFA (MGCFA), is the most widely used method for evaluating MI (e.g., Cheung and Rensvold 2002; Putnick and Bornstein 2016). It proceeds testing through a hierarchy of invariance levels with increasingly stronger assumptions: *configural invariance* (the same structural pattern holds in all groups), metric invariance (factor loadings are additionally equal, indicating that items relate to the construct with the same strength across groups), *scalar invariance* (item intercepts are additionally equal, establishing a common

scale unit and origin needed for comparing latent means) (Little 1997), and *residual* or *strict invariance* (residual variances are additionally equal, so that any remaining group differences reflect the construct rather than differential measurement error) (Cheung and Rensvold 2002; Widaman and Reise 1997). Which level is required depends on the inference being drawn, e.g., metric invariance is generally necessary for comparing latent means, whereas weaker forms may suffice for comparing correlations (Putnick and Bornstein 2016; Little 1997).

Because MGCFA tests invariance only with respect to a pre-specified grouping variable, it is well suited to settings with theoretically meaningful discrete distinctions, e.g. a set of treatment conditions, demographic categories, or discrete measurement occasions. It is less well suited to characteristics that vary continuously, such as age. Forcing these into predefined groups requires arbitrary cutoffs that are hard to justify and can obscure both *where* non-invariance occurs and *how* measurement parameters actually vary along the underlying continuum. Methods that can represent such heterogeneity without requiring either discrete groups or a prespecified functional form are therefore much needed.

Moderated Nonlinear Factor Analysis

MNLFA meets this requirement by modeling factor loadings, intercepts, and other measurement parameters as functions of observed covariates, including continuous characteristics (Bauer and Hussong 2009; Bauer 2017). In this way, MNLFA extends conventional group-based invariance testing by representing measurement heterogeneity without imposing arbitrary group boundaries. MNI is conceptualized as moderation of the measurement model itself by variables exogenous to the model. If measurement parameters differ as a function of these moderator variables, the measurement model no longer operates equivalently across individuals and therefore violates MI (Bauer 2017; Kolbe et al. 2024).

Hence, whether MI is present depends on which parameters are moderated by the exogenous variables. The exogenous variables may moderate any subset of the parameters in the measurement model and structural model. Specifically, the means, variances, and

covariance of latent factors, as well as the item intercepts, factor loadings, and residual variances of any manifest indicator, may vary systematically across individuals as deterministic functions of the exogenous variable (Bauer and Hussong 2009; Bauer 2017). If moderation is limited to the latent-factor parameters, that is, the means, variances, and covariance, the model remains consistent with MI. In contrast, moderation of item-level parameters, including intercepts, loadings, or residual variances, constitutes MNI (Bauer 2017; Kolbe et al. 2024). Accordingly, a moderation function must be specified for each parameter that is allowed to vary as a function of the exogenous variable. Because of this conceptualization,this method allows researchers to model gradual, nonlinear, and individual-specific forms of MNI that are difficult to represent within traditional multiple-group approaches (Bauer and Hussong 2009; Kolbe et al. 2024).

This increased flexibility comes at the cost of greater specification requirements. Like MGCFA, MNLFA constitutes a confirmatory approach to the assessment of MI (Bauer 2017). In practice, at least three sets of decisions are required. First, the measurement model must be specified, including the latent structure and the relationships between latent and observed variables. Second, one or more moderator variables must be selected to represent potential sources of heterogeneity in the measurement process. Third, moderation functions must be specified for all parameters that are allowed to vary as a function of the moderator. This includes determining which parameters are moderated as well as specifying the functional form of each moderation effect (e.g., linear, quadratic, or another parametric form). Consequently, the detection of MNI depends not only on the choice of moderator variables but also on the assumptions made about how these moderators influence the model parameters (Bauer and Hussong 2009; Bauer 2017).

This specification becomes challenging when there is limited prior knowledge about which candidate covariates are associated with measurement parameters, which parameters exhibit heterogeneity, or what functional forms characterize these relationships. Such

settings motivate methods that can explore parameter heterogeneity without requiring these moderation relationships to be fully specified a priori.

Structural Equation Model Trees

SEM trees were originally introduced by Brandmaier et al. (2013) as a method for identifying important covariates and their interactions within SEMs to explain heterogeneity, e.g. by identifying moderators of model parameters in a SEM. Subsequent developments further refined the methodology and expanded its applicability (Arnold et al. 2021; Brandmaier et al. 2016).

Notably, SEM Trees were not developed as a dedicated methods to investigate MI but as a generic strategy for explaining heterogeneity. Because violations of MI manifest as systematic heterogeneity in parameters of the measurement model, SEM trees are by design well suited for detecting such effects. Even though SEM trees have received relatively little attention as tools for detecting MNI in general, some researchers have already used SEM Trees as a successful heuristic to investigate MI (Finch 2017; Sajobi et al. 2026).

SEM trees merge two paradigms: SEMs and decision trees. Decision trees describe recursive partitions of the covariate space, whereby the sample is repeatedly split according to covariate values into progressively more homogeneous subsets, in order to maximize differences in a given outcome variable. SEM trees combine the hierarchical structure of decision trees with SEMs, so that they describe heterogeneity in data with respect to a specified SEM as targeted outcome structure. Unlike many other tree-based methods, SEM trees can be applied to models that include latent variables.

In contrast to MGCFA and MNLFA, SEM trees represent a rather exploratory approach to the assessment of MI. Similar to these methods though, SEM trees require the researcher to specify a measurement model a priori. Beyond this, however, the researcher only needs to provide a set of candidate variables that may account for heterogeneity in the

data. These variables may be numerous and can be either categorical or continuous.

Similar to MGCFA, researchers first specify a measurement model and identify variables that may account for heterogeneity. The crucial difference is that MGCFA requires the researcher to select a specific categorical grouping variable, whereas SEM trees can evaluate a large set of candidate variables simultaneously and determine which, if any, are associated with violations of MI. Moreover, SEM trees can accommodate heterogeneity generated by nonlinear forms of moderation without requiring the functional form to be specified parametrically, by representing such heterogeneity through recursive partitioning.

Hence, SEM trees combine a theory-driven measurement model with a data-driven search for heterogeneity. This makes them especially valuable when researchers have a well-established measurement model but little prior knowledge about where MNI may arise or what form it may take. Because the decision structure closely resembles familiar MGCFA practice, SEM trees represent a comparatively accessible extension to conventional MI analysis.

Present Study

MGCFA, MNLFA, and SEM trees represent measurement heterogeneity differently. MGCFA captures it as differences between predefined groups. When heterogeneity instead depends on a continuous covariate z, the problem becomes modeling a measurement parameter θ as a function $\theta(z)$: MNLFA specifies this function parametrically, whereas SEM trees approximate it through a data-driven, piecewise linear approximation.

The present simulation examines the consequences of this difference for detecting MNI, comparing MNLFA and SEM trees when the true $\theta(z)$ relationship is linear, quadratic, or sigmoidal, and varying the magnitude of heterogeneity and sample size. Three aspects of performance are addressed: whether each method detects MNI at all; how detection depends on the functional form of $\theta(z)$; and, since loading and intercept heterogeneity are introduced separately, whether each method correctly localizes MNI to the metric or scalar level.

We expect SEM trees to hold an advantage under nonlinear, especially sigmoidal, moderation, with this advantage shrinking under purely linear moderation, which is typically more difficult to represent well in the recursive partitioning paradigm. MNLFA, conversely, should perform best when its specified functional form is correct — particularly under linear moderation — but be more sensitive to misspecification. For both methods, stronger moderation effects and larger sample sizes should improve detectability.

We complement the simulation with an empirical illustration using CASP-12 quality-of-life data from Wave 9 of the Survey of Health, Ageing and Retirement in Europe (SHARE-ERIC 2024; Borrat-Besson et al. 2015; Wiggins et al. 2008), motivated by Bonneville-Roussy et al. (2024)'s group-based analysis of CASP-12 invariance across age, gender, and cultural region. Age offers a natural comparison case: MGCFA requires discretizing it into groups, whereas MNLFA and SEM trees can treat it as continuous, avoiding arbitrary categorization.

We close with an integrative discussion of the simulation and empirical findings and practical recommendations for when MGCFA, MNLFA, or SEM trees are best suited to testing measurement invariance.

Simulation

Method

To compare MNLFA and SEM trees under conditions where the true relationship between a continuous covariate and measurement parameters is known, we ran a Monte Carlo simulation study. Data were generated from a single-factor measurement model in which factor loadings and/or item intercepts were made to depend on a continuous moderator through one of several functional forms (linear, quadratic, sigmoidal, or none). We systematically varied the functional form of moderation, the parameter class affected (loadings, intercepts, or both), the magnitude of moderation, sample size, and analysis method. For each combination of these design factors, a separate dataset was generated and

analyzed using the corresponding method: linear MNLFA, quadratic MNLFA, or SEM trees. Thus, conditions differing only in analysis method were based on independently generated datasets under otherwise identical data-generating conditions. This design allowed us to evaluate, under fully controlled conditions, whether each method detected the presence of MNI, how detection was affected by a match or mismatch between the true and assumed functional form, and whether each method correctly localized detected heterogeneity to the metric or scalar level.

Model Specification

In the CFA, baseline factor loadings were fixed to $\lambda = 0.70$, baseline intercepts were fixed to one, and the latent factor variance was fixed to one.

Residual variances were fixed such that indicator reliability equaled $\rho = .75$. at the baseline factor loading of $\lambda = .70$. Because residual variances were held constant while loadings varied under loading moderation, conditional indicator reliability varied accordingly in conditions with metric noninvariance. MNI was introduced by allowing selected factor loadings and/or item intercepts to depend on the moderator.

Nine population models were investigated.

MNI was generated by allowing selected measurement parameters to vary continuously as deterministic functions of an observed moderator variable. For each simulated individual, a moderator variable $M \sim \mathcal{U}(-1, 1)$ was sampled from a continuous uniform distribution and transformed into the moderator $Z = h(M)$, where the transformation $h(\cdot)$ determined the functional form of moderation. Measurement parameters were subsequently modeled as linear functions of the transformed moderator. Consequently, any nonlinearity in parameter variation arose exclusively through the transformation $h(\cdot)$.

Functional forms of moderation

Four moderator functions were considered (see Figure 1 for an illustration):

Table 1

Overview of the data generating population-models.

Model	Loading moderation	Intercept moderation
L0I0	none	none
LAI0	all items	none
L0IA	none	all items
LAIA	all items	all items
L2I0	items 1–2	none
L0I2	none	items 1–2
L2I2	items 1–2	items 1–2
1.3	all items	none
1.32	all by M1	all by M2

Note. L = loading; I = intercept; 0 = no moderation; 2 = moderation of items 1–2; A = moderation of all items.

1. **Linear:**

$$h(M) = M$$

In this case, the transformed moderator is bounded to the interval $(-1, 1)$ and rotation symmetric around zero.

2. **Sigmoid:**

$$h(M) = a + \frac{b - a}{1 + \exp\Big(-k(M - c)\Big)}, \qquad a = -1,\ b = 1,\ c = 0,\ k > 0.$$

This transformation maps the input to the open interval $(-1, 1)$ and yields a bounded moderator, rotation symmetric around zero, exhibiting diminishing sensitivity at extreme values. In our simulation k is set to 10.

3. **Quadratic:**

$$h(M) = 2M^2 - 1$$

This transformation captures symmetric nonlinear moderation as a function of distance from the center while remaining bounded in $(-1, 1)$.

4. **Null:**

$$h(M) = 0$$

These functions were selected to represent qualitatively different forms of heterogeneity. The first three functions represented structured forms of moderation, whereas the null condition represented the absence of systematic moderation. All moderator transformations were bounded to the interval $(-1, 1)$, implying identical maximum parameter deviations for a given moderation coefficient, but not necessarily the same variance. Consequently, identical values of the moderation coefficient imply identical maximum parameter deviations but not necessarily identical average parameter variation.

Simulation design

The simulation followed a fully crossed factorial design, combining data-generating conditions with analysis method. Data-generating conditions varied the population measurement model (whether and which parameters, that is, loadings, intercepts, or both, were subject to moderation), the functional form of that moderation (linear, quadratic, sigmoidal, or null), the magnitude of loading and intercept moderation, and sample size. For each combination of data-generating conditions and analysis method, a separate dataset was generated and analyzed using the method specified for that condition, linear MNLFA, quadratic MNLFA, or SEM trees. To keep track of the different population models, we introduced a four-letter coding scheme (see table 1). Each code first indicates whether factor loadings were moderated, no loading moderation is denoted L0, moderation of two of the four loadings as L2, and moderation of all four loadings as LA. The code then indicates whether item intercepts were moderated, following the same logic: no intercept moderation is denoted

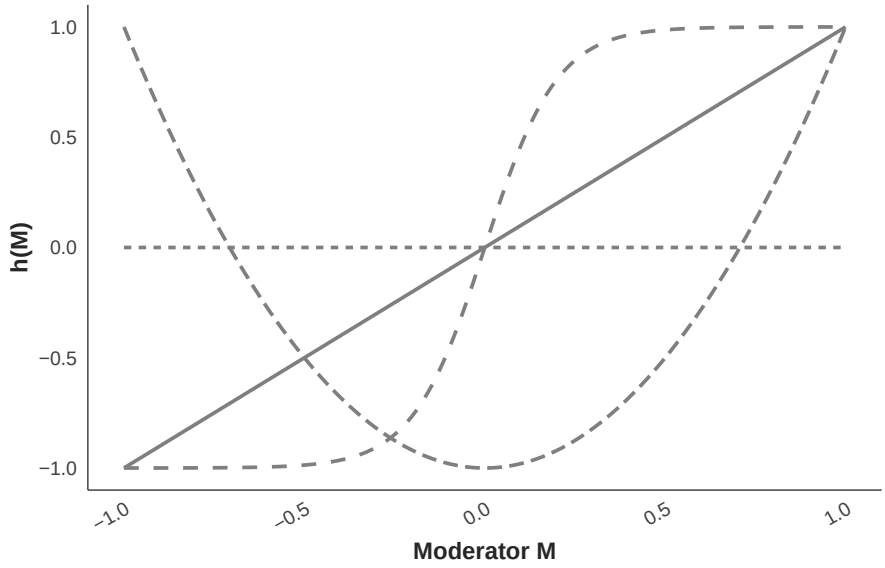


Figure 1

Note. The figure illustrates the linear (solid line), quadratic (long-dashed line), and sigmoidal (short-dashed curve) moderator functions considered in the simulation study. The horizontal dotted line represents the absence of a moderator effect. The functions are shown over the moderator range from -1 *to* 1*, with* $h(M)$ *representing the moderator effect on a measurement parameter.*

I0, moderation of two of the four intercepts as I2, and moderation of all four intercepts as IA. For example, a population model in which all loadings but no intercepts were moderated is denoted LAI0. Table 2 summarizes the full set of manipulated factors and their levels.

Primary and exploratory simulation conditions

Accounting for the applicability of the manipulated factors across population models and functional forms, the simulation design comprised 508 distinct data-generating cells. Each condition was evaluated using linear MNLFA, quadratic MNLFA, and SEM trees across 100 replications, yielding 152,400 primary analyses.

Inspection of the primary results indicated that several conditions yielded near-ceiling detection rates, limiting differentiation between methods at the preregistered effect sizes.

Table 2

Overview of the simulation design.

Factor	Levels
Population model	L0I0, LAI0, L0IA, LAIA, L2I0, L0I2, L2I2, 1.3, 1.32
Functional form of moderation	Linear, quadratic, sigmoid, null/no moderation
Loading moderation magnitude	$\delta_\lambda \in \{0.20, 0.30\}$
Intercept moderation magnitude	$\delta_\nu \in \{0.50, 1.00\}$
Sample size	$N \in \{300, 500, 700, 1000\}$
Indicator reliability	$\rho = .75$
Baseline factor loading	$\lambda = .70$
Baseline item intercept	$\nu = 1$
Analysis method	Linear MNLFA, quadratic MNLFA, SEM tree

Note. MNLFA = moderated nonlinear factor analysis; SEM = structural equation modeling; δ_λ = magnitude of loading moderation; δ_ν = magnitude of intercept moderation; N = sample size; ρ = indicator reliability. Population-model codes are explained in Table 1.

Exploratory analyses therefore additionally examined smaller moderation effects ($\delta_\lambda = .10, \delta_\nu = .25$).

In the primary method comparison, the relevant data-generating moderator or moderators were supplied to the SEM trees as candidate partitioning variables. Thus, the comparison focused on how the methods represent moderator-related parameter heterogeneity. Candidate-variable selection was examined separately in a robustness analysis.

Analysis Models

The simulated population conditions were evaluated using two parametrically specified MNLFA analyses, linear MNLFA and quadratic MNLFA, as well as score-based SEM trees. All analyses were based on a single-factor measurement model with four continuous indicators $x_1, \ldots, x_4$ and one latent factor η. The MNLFA analyses evaluated

metric and scalar MNI through nested likelihood-ratio tests, whereas the SEM-tree analyses used separate score-based parameter-instability tests focused on factor loadings and item intercepts.

MNLFA

MNLFA models were estimated in `OpenMx` using raw-data maximum likelihood. To identify the model, the latent-factor mean and variance were fixed to 0 and 1, respectively. Baseline item intercepts, factor loadings, and residual variances were freely estimated.

Let M_1 and M_2 denote the raw moderator variables. In the data-generating model, the transformed moderators were defined as $Z_k^{(d)} = h(M_k)$, where $h(\cdot)$ denotes the data-generating transformation. In the fitted MNLFA, the transformed moderators were analogously defined as $Z_k^{(a)} = g(M_k)$, where $g(\cdot)$ denotes the transformation assumed in the analytical model. For conditions involving two moderators, the interaction term was defined as $Z_{12}^{(a)} = Z_1^{(a)} Z_2^{(a)}$. In the unrestricted MNLFA, the intercept of item i was specified as

$$\nu_i \left(Z_1^{(a)}, Z_2^{(a)} \right) = \nu_{i0} + \beta_{\nu i1} Z_1^{(a)} + \beta_{\nu i2} Z_2^{(a)} + \beta_{\nu i12} Z_{12}^{(a)},$$

and its factor loading as

$$\lambda_i \left(Z_1^{(a)}, Z_2^{(a)} \right) = \lambda_{i0} + \beta_{\lambda i1} Z_1^{(a)} + \beta_{\lambda i2} Z_2^{(a)} + \beta_{\lambda i12} Z_{12}^{(a)}.$$

Thus, moderator effects associated with $Z_1^{(a)}$, $Z_2^{(a)}$, and their interaction $Z_{12}^{(a)}$ were freely estimated for every item intercept and factor loading. Residual variances were additionally allowed to vary with $Z_1^{(a)}$ and $Z_2^{(a)}$ according to the log-linear specification

$$\theta_i \left(Z_1^{(a)}, Z_2^{(a)} \right) = \theta_{i0} \exp \left(\gamma_{i1} Z_1^{(a)} + \gamma_{i2} Z_2^{(a)} \right).$$

Residual-variance moderation was included as a nuisance component of the fitted MNLFA and was not used to define metric or scalar MNI. No interaction term was included for residual-variance moderation. The latent mean was fixed at zero and was not moderated, and the latent variance was fixed at one.

Metric MNI was evaluated by comparing the unrestricted model with a metric-invariant model in which all moderator effects on the factor loadings were constrained to zero. Moderator effects on item intercepts and residual variances remained freely estimated. The resulting likelihood-ratio test[1] therefore evaluated the joint null hypothesis that no factor loading varied as a function of the modeled moderators. Scalar MNI was evaluated by comparing the metric-invariance model with a scalar-invariant model in which moderator effects on item intercepts were additionally constrained to zero. Residual-variance moderation remained freely estimated. The scalar likelihood-ratio test therefore evaluated the joint null hypothesis of no moderator-related intercept variation conditional on loading moderation having been constrained to zero. An additional omnibus comparison between the unrestricted and scalar-invariance models tested the joint removal of loading and intercept moderation. All likelihood-ratio tests used $\alpha = .05$.

The linear and quadratic MNLFA specifications differed only in the transformation $g(\cdot)$ applied to the raw moderators. Linear MNLFA used $g(M) = M$, whereas quadratic MNLFA used $g(M) = 2M^2 - 1$. Thus, quadratic MNLFA did not represent a conventional polynomial model containing both linear and squared moderator terms; rather, the quadratically transformed moderators entered otherwise identical moderation equations.

The analytical functional form was correctly specified when $g(\cdot) = h(\cdot)$. Accordingly, linear MNLFA matched the linear DGM and quadratic MNLFA matched the quadratic DGM. Because no sigmoid MNLFA was fitted, both specifications were functionally

[1] The degrees-of-freedom of the test correspond to the number of items in the model.

misspecified under sigmoid data-generating conditions.

Score-based SEM trees

For the SEM-tree analyses, a single-factor CFA was specified with all four factor loadings $\lambda_1, \ldots, \lambda_4$, all four item intercepts $\nu_1, \ldots, \nu_4$, and all four residual variances freely estimated. The latent-factor mean and variance were fixed at 0 and 1, respectively. This CFA served as the base model for growing the SEM tree to detect parameter heterogeneity (also known as parameter instability in the score-based testing literature) in the measurement model.

Separate SEM trees were estimated for loading-related and intercept-related parameter heterogeneity. In the *metric tree*, the four factor-loading parameters $\lambda_1, \ldots, \lambda_4$ were specified as focus parameters in the score-based instability tests. Candidate splits were therefore evaluated with respect to instability in the loading parameters only. In the *scalar tree*, the four item-intercept parameters $\nu_1, \ldots, \nu_4$ were instead specified as focus parameters, such that the instability test targeted heterogeneity in the intercept parameters.

Importantly, focus parameters determined which parameter scores entered the instability test; non-focused free parameters were allowed to vary between groups but did not contribute to the split evaluation. Thus, item intercepts and residual variances remained free parameters of the CFA underlying the loading-focused tree, whereas factor loadings and residual variances remained free parameters of the CFA underlying the intercept-focused tree. The metric and scalar SEM trees therefore constituted separate parameter-focused instability analyses rather than a sequence of CFA models in which increasingly restrictive equality constraints were imposed on the complete measurement model. This implementation detail is particularly relevant for interpreting cross-level rejection, because instability detected by a loading-focused or intercept-focused test need not imply that heterogeneity is confined exclusively to the corresponding parameter class.

Recursive partitioning used score-based parameter-instability tests with $\alpha = .05$, Bonferroni adjustment for multiple testing, a maximum tree depth of three, and a minimum node size of 50.

In the primary comparison, SEM trees were supplied with the relevant raw data-generating moderator M_1, or moderators M_1 and M_2, as candidate partitioning variables. Unlike MNLFA, the SEM-tree analyses were therefore not supplied with the transformed moderators $Z^{(d)} = h(M)$. Under quadratic and sigmoid data-generating conditions, the tree instead had to represent the resulting parameter heterogeneity through recursive partitions of the original moderator space M.

Candidate-variable selection was examined separately in a robustness analysis. In this condition, the true raw moderator M_1 was supplemented with ten independent irrelevant predictors, each generated independently from a uniform distribution on $(-1, 1)$. The robustness analysis was restricted to the all-item metric-MNI condition with linear moderation, $N = 500$, $\delta_\lambda = .20$, and $\delta_\nu = .50$. It therefore evaluated whether the SEM tree could select the informative moderator when irrelevant candidate partitioning variables were also available.

Evaluation criteria

Performance was evaluated separately for metric and scalar MNI. For MNLFA, metric and scalar MNI were assessed using sequential likelihood-ratio tests comparing nested moderation models. For SEM trees, MNI was assessed using score-based global parameter-instability tests on the respective focus parameters (factor loadings or intercepts) with Bonferroni-adjusted significance tests (Arnold et al. 2021; Merkle and Zeileis 2013).

For each population model and simulation condition, the following performance measures were computed:

Statistical power: The proportion of simulation replications in which the method

rejected the null hypothesis of MI at the corresponding level when the respective form of MNI was present in the data-generating model.

Complete-null rejection: The proportion of simulation replications in which the method rejected the null hypothesis of MI at the corresponding level under complete MI, when neither loading nor intercept MNI was present.

Level-specific rejection: The proportion of simulation replications in which the method rejected the null hypothesis of MI at the corresponding level when that form of MNI was absent, but MNI could be present at another level. For example, metric rejection under scalar-only MNI was considered level-specific rejection rather than rejection under complete MI.

In addition, a *sequential classification* was derived using a metric-first decision rule, following traditional approaches to MI testing. If the metric test rejected, the replication was classified as metric MNI; if the metric test did not reject but the scalar test rejected, it was classified as scalar MNI; if neither test rejected, invariance was retained. In the primary comparison, SEM trees were supplied only with the true moderator variables. Consequently, statistical power and rejection rates reflect the ability to detect MNI independent of moderator selection.

In the robustness analysis a simulation replication was considered successful only if the SEM tree both rejected the null hypothesis of parameter stability and selected the true moderator as a splitting variable. This criterion therefore jointly evaluated the ability of SEM trees to detect MNI and to distinguish the informative moderator from irrelevant candidate predictors.

Exploratory Analysis

After seeing the first simulation results, recognizing that we obtained effects from a rather uninformative, or homogeneously informative space, we appended runs with smaller

effect sizes for δ_{lambda} (.10) and δ_{nu} (.25). The respective results are to be understood as exploratory results, diverging from the preregistrated delta values.

Replications

Each data-generating design cell was replicated 100 times. The three analysis methods were evaluated in separate method-specific simulation runs. Method-specific replications used independently seeded datasets, and performance was therefore compared at the simulation-condition level rather than through paired replication-level contrasts. An additional 100 SEM-tree analyses were conducted for the multiple predictor robustness check.

Transparency and Openness

All data generating code, analysis code, and research materials are available at [https://github.com/LeonieHagitte/SEMi]. This study's design and its analysis were pre-registered, following the ADEMP preregistration template (Siepe et al. 2024), at Zenodo (10.5281/zenodo.19891929). For the empirical example we used the ninth wave data of SHARE, which is not openly available, but can be requested for scientific use. The questionnaires referred to are openly available online.

Software

All simulations and data analyses were conducted in R (R Core Team 2025). MNLFA models were estimated using `OpenMx` (Boker et al. 2026) with raw-data maximum likelihood and definition variables. Score-based SEM trees were estimated using the `semtree` package (Brandmaier et al. 2026). Simulation design construction, data handling, aggregation, and visualization did rely on `dplyr`(Wickham et al. 2026), `tidyr` (Wickham et al. 2025), `tibble` (Müller and Wickham 2026), `purrr` (Wickham and Henry 2026), `future` (Bengtsson 2021), `parallel` (R Core Team 2025), and `ggplot2` (Wickham 2016).

Results

Across 152,400 method-specific simulation runs, 42 (0.028%) did not yield evaluable metric or scalar test results.

This occurred in 8 of 50,800 MNLFA runs (0.016%) and 34 of 50,800 MNLFAQ runs (0.067%); all 50,800 SEM tree runs yielded evaluable test results.

Whenever results were unavailable, both metric and scalar test results were missing.

Rejection rates were calculated based on the number of evaluable replications within each condition, with all conditions comprising 100 attempted replications.

Under complete MI (Model L0I0), rejection rates were generally close to the nominal $\alpha = .05$ level. This pattern was observed across the three methods and was comparatively stable across the examined sample sizes and moderator functional forms. Complete condition-specific rejection rates, including Monte Carlo standard errors, are reported in Supplementary Table S1.

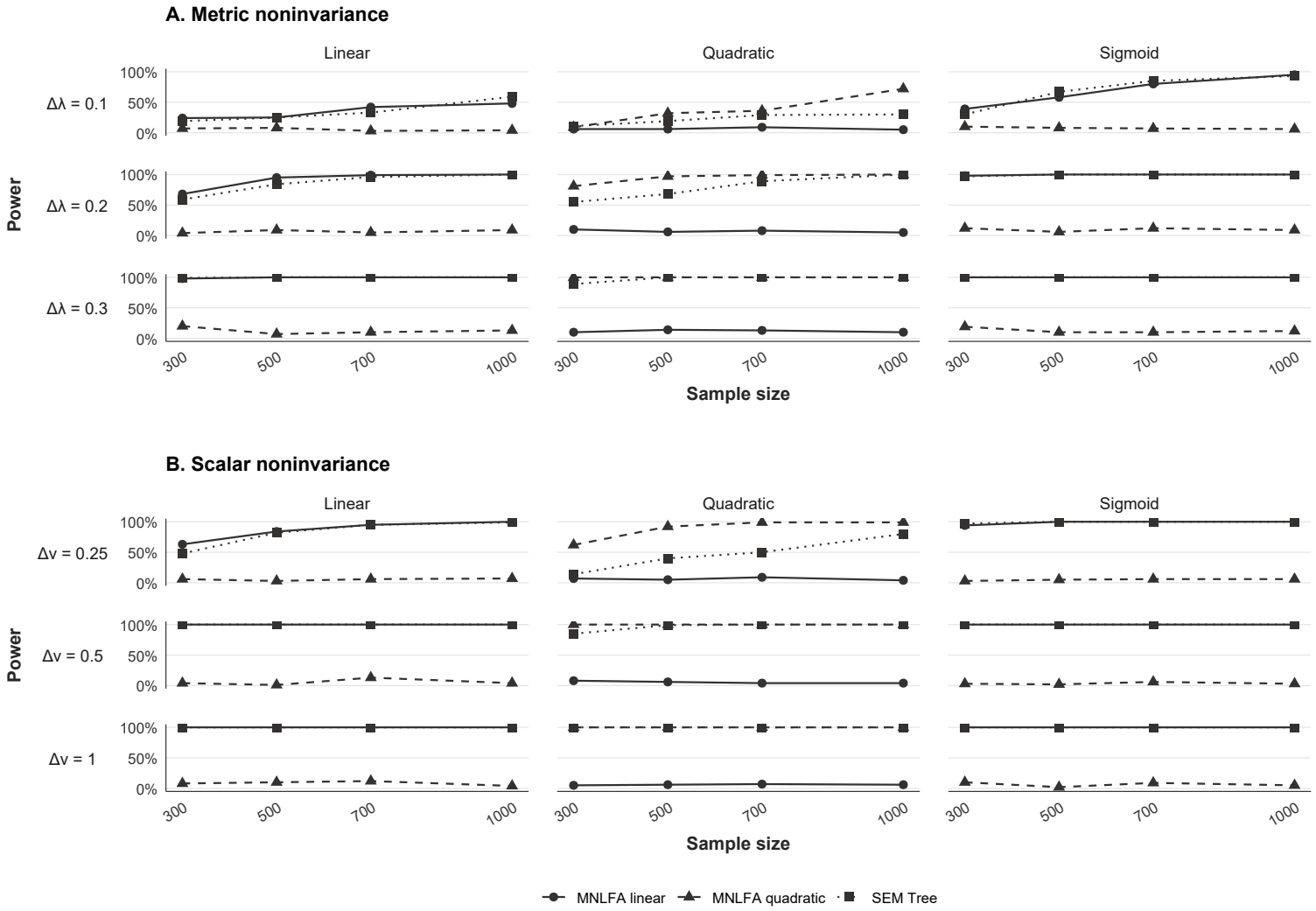


Figure 2

Note. Statistical power across sample sizes, moderator functional forms, and effect magnitudes for linear MNLFA, quadratic MNLFA, and SEM trees. Panel A shows metric power under metric-only noninvariance (Model LAI0), with $\Delta_\nu = 0.5$*; Panel B shows scalar power under scalar-only noninvariance (Model L0IA), with* $\Delta_\lambda = 0.2$*. Effect magnitudes of* $\Delta_\lambda = 0.1$ *and* $\Delta_\nu = 0.25$ *represent exploratory simulation conditions.*

Statistical power depended strongly on the correspondence between the functional form of MNI and the moderator specification used in MNLFA. Linear MNLFA generally showed greater sensitivity to linear and sigmoid heterogeneity than to quadratic heterogeneity. Conversely, quadratic MNLFA performed particularly well under quadratic moderation, whereas its sensitivity under linear and sigmoid moderation was substantially lower in several conditions. SEM trees showed comparatively consistent detection performance across moderator functional forms, although their power also varied with sample size, effect magnitude, and the population-model configuration. Figure 2 illustrates these patterns for metric-only noninvariance (Model LAI0) and scalar-only noninvariance (Model L0IA).

Power generally increased with sample size, particularly for SEM trees, which often approached ceiling performance by $N = 700$ or $N = 1000$. However, increasing sample size did not compensate for functional-form mismatch in MNLFA. The linear specification remained weak under quadratic MNI even at larger sample sizes, whereas the quadratic specification remained weak under linear and sigmoid MNI. Beyond functional form and sample size, power also varied across MNI configurations. SEM tree metric power was consistently high across most population models but was comparatively lower for Model 1.32, in which loading and intercept noninvariance were induced by different moderators. The patterns shown in figure 2 represent selected population models and explicitly defined effect-size conditions. The complete results across population models, sample sizes, moderator forms, and confirmatory and exploratory effect magnitudes are reported in Supplementary Table S2. Thus, although SEM trees were generally robust across functional forms, their performance also depended on the specific configuration of MNI.

Rejection behavior differed when invariance held at the level being tested but was violated at another measurement level. In particular, under scalar-only noninvariance, metric-level rejection rates sometimes substantially exceeded the nominal significance level

despite the absence of loading heterogeneity. This was particularly evident for SEM trees and for the functionally mismatched MNLFA specification.

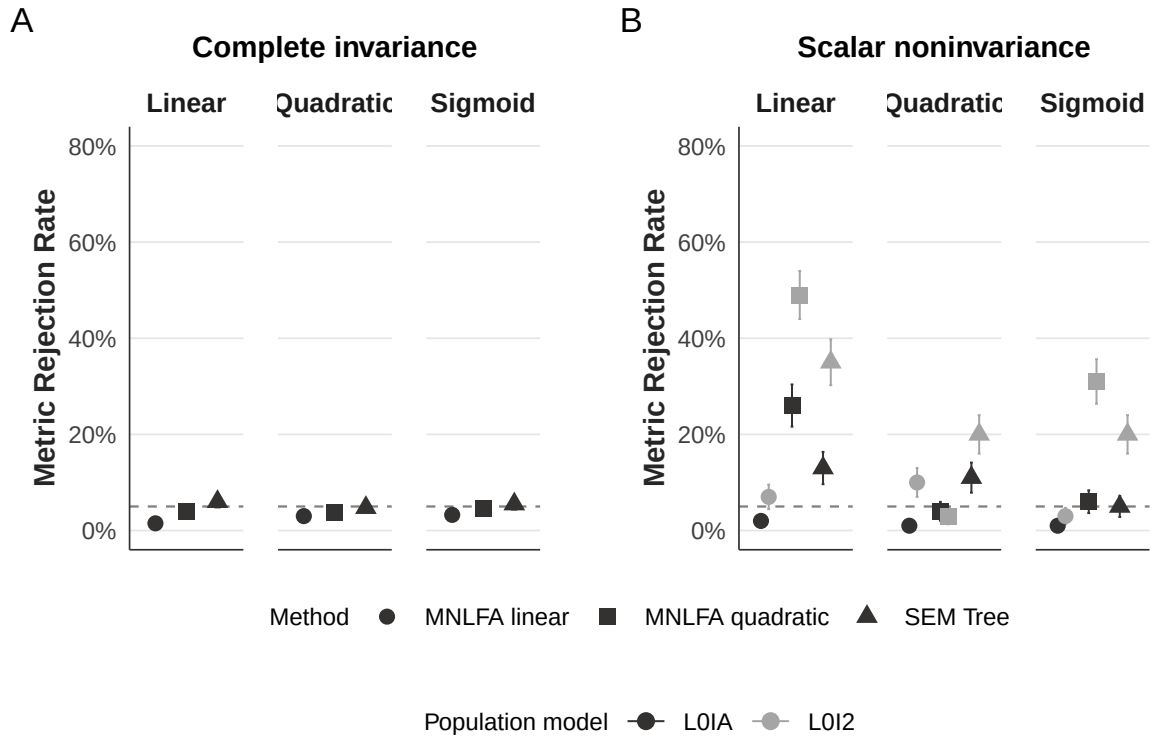


Figure 3

Note. Metric-level rejection rates under complete MI (A) and scalar-only noninvariance (B), by moderator functional form and analysis method, at N = 1000 (δ_λ = 0.2. & δ_ν = 0.5) The dashed horizontal line indicates the nominal significance level of $\alpha = .05$. Elevated metric rejection under scalar-only noninvariance represents rejection of a true metric-invariance hypothesis in the presence of intercept heterogeneity.

3 contrasts metric rejection under complete invariance with metric rejection in the presence of scalar-only noninvariance. Whereas rejection rates under complete invariance were generally near the nominal level, intercept heterogeneity could produce substantially elevated metric rejection rates. This indicates cross-level sensitivity rather than complete-null Type I error and high sensitivity of the SEM tree metric test was accompanied

by reduced specificity for the level of noninvariance when scalar MNI was present. The elevated metric rejection rates in the presence of scalar MNI also varied with sample size and number of moderated intercepts. In several conditions, particularly for SEM trees, metric rejection increased as sample size increased, potentially indicating that cross-level sensitivity became more pronounced with increasing statistical information. In contrast, scalar rejection rates when intercepts were invariant remained generally close to the nominal level across methods and functional forms, indicating considerably less cross-level rejection of scalar tests in the presence of loading MNI

This finding has implications beyond the interpretation of individual invariance tests. In a sequential testing procedure, rejection at the metric stage prevents subsequent evaluation of scalar invariance. Consequently, elevated metric rejection under scalar-only noninvariance can lead to an incorrect final classification even when the method would otherwise be sensitive to intercept heterogeneity.

Figure 4 presents correct classification rates at ($N = 1000$) under complete invariance, metric-only noninvariance, and scalar-only noninvariance. Classification performance varied across methods, moderator functional forms, and importantly, effect magnitudes. Under scalar-only noninvariance, the results were particularly sensitive to the behavior of the preceding metric test, when the effect magnitude was highest. The proportion of replications reaching the scalar stage provides a possible explanation for this pattern. At $N = 1000$ ($\Delta_\lambda = 0.2$, $\Delta_\nu = 1.0$) linear MNLFA proceeded to scalar testing in 95%, 94%, and 94% of replications under linear, quadratic, and sigmoid moderation, respectively. The corresponding proportions were 0%, 93%, and 45% for quadratic MNLFA and 4%, 22%, and 88% for SEM trees. Thus, in several conditions, a substantial proportion of replications had already been incorrectly classified as exhibiting metric noninvariance, preventing the scalar test from being reached. This distinction is essential when comparing level-specific power with the performance of the complete sequential procedure. Complete condition-specific

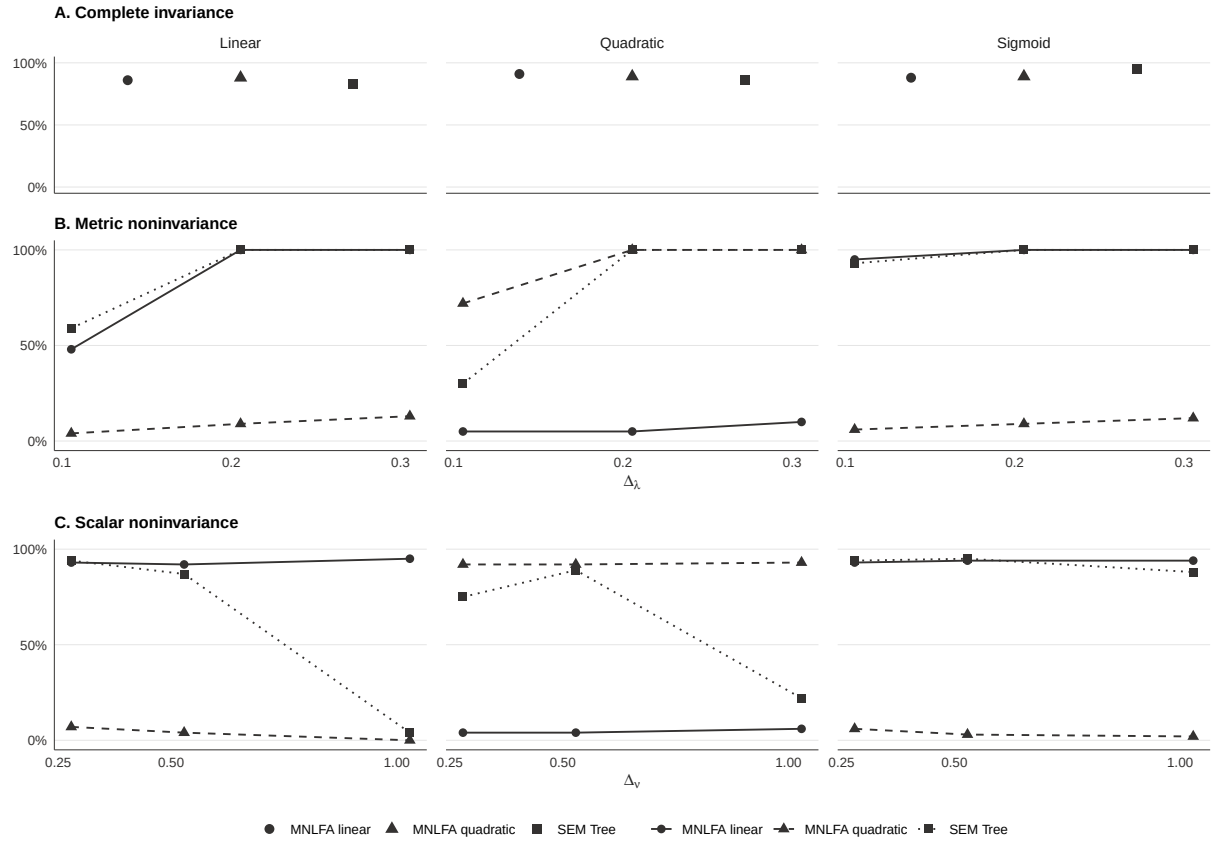


Figure 4

Note. Correct classification rates under sequential metric-then-scalar invariance testing at $N = 1000$*, by moderator functional form, effect magnitude, and method. Panels show complete invariance (A), metric-only noninvariance (B), and scalar-only noninvariance (C). Scalar testing was conducted sequentially only when metric invariance was retained. Under scalar-only noninvariance, elevated metric-level rejection rates therefore reduced the number of replications reaching scalar testing. At* $N = 1000$ *and* $\Delta_\nu = 1.0$*, this proportion ranged from 4% to 88% for SEM trees and from 0% to 93% for quadratic MNLFA across moderator forms, compared with 94% to 95% for linear MNLFA.*

probabilities of metric classification, scalar classification, retained invariance, and correct final classification are provided in Supplementary Table S3.

In the candidate-variable robustness analysis (1,000 replications), metric non invariance was detected in 79.1% of replications and the true moderator was identified in 77.4%. The true moderator alone was selected in 68.2% of replications, while the true moderator and at least one noise variable were jointly selected in 9.2%. Selection of noise variables in the absence of the true moderator occured in 1.7% of the cases.

Illustrative Application: Quality of Life (CASP-12)

We illustrate both approaches using CASP-12 data from Wave 9 of the Survey of Health, Ageing and Retirement in Europe (SHARE). The CASP-12, a short form of the CASP-19 quality-of-life measure (Hyde et al. 2003), comprises 12 items spanning four domains (Control, Autonomy, Self-Realization, and Pleasure) each measured by three items on a four-point scale (1 = Often to 4 = Never). The SHARE version deviates from the standard 12-item form proposed by Wiggins et al. (2008) in two items (A2, P3)[2] Positively worded items were reverse-scored so that higher values consistently indicated better quality of life; items were treated as continuous indicators, so scalar parameters correspond to item intercepts.

Following Bonneville-Roussy et al. (2024), we considered age (continuous), gender (dichotomous), and cultural region (Nordic, Western, Southern, Eastern Europe; Western as reference) simultaneously as candidate sources of measurement heterogeneity. Gender was coded as 0 = male and 1 = female, and Western Europe served as the reference category for cultural region in the MNLFA analyses.

[2] SHARE replaces “I feel that I can please myself what I do” with an item on feeling hindered by family obligations (A2), and “I enjoy the things that I do” with an item on looking back on one’s life with happiness (P3).

Both analyses used the first item of each factor (C1, A1, P1, S1) as anchors, with loadings fixed to 1 and intercepts and moderation effects fixed to 0, so that MNI was tested for the remaining eight items conditional on anchor invariance. In the MNLFA, the loadings, intercepts, and residual variances of the eight non-anchor items were allowed to vary with all three covariates; latent means were likewise freely moderated across the metric and scalar models, separating covariate-related mean differences from intercept moderation, while the latent covariance matrix was estimated but held constant. The SEM-tree analyses estimated separate loading- and intercept-focused trees on the same eight non-anchor items, with age, gender, and cultural region supplied jointly as candidate partitioning variables and the anchor items excluded from the focal parameters. As in the simulation, score-based tests used $\alpha = .05$ with Bonferroni adjustment and a maximum depth of three, but with a minimum node size of 100. A SEM forest (Brandmaier et al. 2016) was additionally estimated to obtain partial dependence plots approximating the functional form of moderator-related parameter variation.

CASP-12 responses outside the valid range of 1–4 were coded as missing, as were invalid negative age values. Participants were excluded if they had missing information on age, gender, or cultural region. For the MNLFA and SEM-tree analyses, the sample was further restricted to participants with complete responses on all 12 CASP-12 items. Age was standardized within the resulting analysis sample of $N = 65{,}214$ participants.

Results

First, we performed a CFA, using WLSMV as estimator, to assess global model fit. The simple CFA model for CASP-12 ($X^2 = 12{,}532.79$; $df = 48$) showed acceptable fit indices with CFI = 0.99, TLI = 0.98, RMSEA = 0.063 [0.062–0.064], and SRMR = 0.05. The loadings onto the items were generally high only the factor *Autonomy* showed lower loadings onto its items A1 ($\lambda_{A1} = 0.58$), A2 ($\lambda_{A2} = 0.25$), and A3 ($\lambda_{A3} = 0.45$). Full results can be found in the Appendix table .

MNLFA

Metric and scalar MNI of the CASP-12 were evaluated through the sequence of nested MNLFA model comparisons described above, testing first whether loading-moderation effects, and then whether intercept-moderation effects, could be constrained to zero without a significant loss of fit.

The unrestricted model provided the best relative fit (-2LL = 1,756,979, AIC = 1,757,383, BIC = 1,759,218).

Constraining loading-moderation effects to zero resulted in a significant deterioration in model fit relative to the unrestricted model $\Delta\chi^2(40) = 2{,}525.23$, p< .001, accompanied by increases in AIC = 1,759,828 and BIC = 1,761,300. Thus, metric invariance was not supported. Subsequently constraining intercept-moderation effects to zero resulted in a further significant deterioration in fit $\Delta\chi^2(40) = 13{,}549.74$, p< .001, with AIC = 1,773,298 and BIC = 1,774,406(see Table 3).

Table 3

MNLFA Model Fit and Measurement Invariance Tests

Model	$-2LL$	AIC	BIC	$\Delta\chi^2$	Δdf	p
Unrestricted	1756978.7	1757382.7	1759218.0	–	–	–
Metric	1759504.0	1759828.0	1761299.8	2525.23	40	$< .001$
Scalar	1773053.7	1773297.7	1774406.1	13549.74	40	$< .001$

Note. MNLFA = moderated nonlinear factor analysis; $-2LL$ = negative twice the log-likelihood; AIC = Akaike information criterion; BIC = Bayesian information criterion. Likelihood-ratio tests compare the metric model with the unrestricted model and the scalar model with the metric model, respectively. Dashes indicate statistics not applicable to the unrestricted model.

Thus, scalar invariance was also not supported. Overall, the model comparisons indicated that both factor loadings and item intercepts varied as a function of the included

moderators.

Inspection of the moderation coefficients suggested that loading non-invariance was particularly associated with cultural cluster and age, whereas gender-related differences were comparatively small (table 4).

Relative to the Western European reference group, the largest positive loading-moderation coefficients for Southern Europe were observed for A3 ($\hat{\beta}_{Southern} = 0.423$) and C3 ($\hat{\beta}_{Southern} = 0.358$)), indicating higher unstandardized factor loadings for these items in Southern Europe. Similarly, the largest coefficients for Eastern Europe were observed for A3 ($\hat{\beta}_{Eastern} = 0.345$) and A2 ($\hat{\beta}_{Eastern} = 0.297$). These estimates indicate differences in the strength of the unstandardized relationships between the respective latent factors and items across cultural clusters. Age-related loading moderation was smaller in magnitude. The largest coefficients were observed for A3 ($\hat{\beta}_{Age}$ = -0.098) and A2 ($\hat{\beta}_{Age}$ = -0.084). Because age was standardized ($SD = 9.74$ years), these coefficients (see table 4) represent changes in the unstandardized factor loadings per one-standard-deviation increase in age, rather than per year. Expressed on the original age scale, the loadings decreased by approximately .010 and .009 units per additional year of age for A3 and A2, respectively, assuming linear moderation and holding the other moderators constant. Thus, the association between the Autonomy factor and these items became weaker with increasing age. Gender-related loading-moderation coefficients ranged from -0.022 to 0.045, indicating that the direction and magnitude of gender-related differences in factor loadings varied across items.

Intercept-moderation coefficients showed larger absolute values than loading-moderation coefficients. The largest coefficients were observed for Southern Europe, particularly for A3 ($\hat{\beta}_{Southern}$ = -1.569) and A2 ($\hat{\beta}_{Southern}$ = -1.220), followed by Eastern Europe for A3 ($\hat{\beta}_{Eastern}$ = -1.247) and A2 ($\hat{\beta}_{Eastern}$ = -0.700)). These negative coefficients indicate lower model-implied item intercepts relative to Western Europe, conditional on the latent factor and the remaining moderators. For example, the intercept of A3 was estimated

Table 4

Item-Level MNLFA Moderation Coefficients

type	item	Age	Eastern	Female	Nordic	Southern
Loading	A2	-0.084	0.297	-0.022	0.001	0.350
Loading	A3	-0.098	0.345	0.017	-0.124	0.423
Loading	C2	0.027	0.165	-0.011	-0.141	0.184
Loading	C3	0.041	0.187	0.045	0.023	0.358
Loading	P2	0.071	0.140	-0.005	-0.126	0.155
Loading	P3	-0.075	0.130	-0.007	-0.155	0.130
Loading	S2	0.028	0.163	0.029	0.055	0.092
Loading	S3	0.024	0.038	0.012	-0.086	0.035
Intercept	A2	0.503	-0.700	0.016	-0.098	-1.220
Intercept	A3	0.621	-1.247	-0.094	0.247	-1.569
Intercept	C2	0.229	-0.257	-0.019	-0.094	-0.291
Intercept	C3	0.124	-0.443	-0.180	-0.383	-0.955
Intercept	P2	-0.265	-0.496	0.010	0.487	-0.429
Intercept	P3	0.299	-0.507	0.020	0.587	-0.316
Intercept	S2	-0.038	-0.631	-0.061	-0.129	-0.492
Intercept	S3	-0.034	-0.238	0.020	0.423	-0.243

Note. Entries are unstandardized moderation coefficients. The Western European cultural cluster and male respondents serve as reference groups. Age coefficients refer to the age scale used in model estimation. Dashes indicate parameters that were not estimated or were constrained for identification.

to be 1.569 units lower in Southern Europe than in Western Europe at equivalent latent factor levels.

Age was also associated with intercept moderation for A3 ($\hat{\beta}_{Age} = 0.621$) and A2 ($\hat{\beta}_{Age} = 0.503$), corresponding to increases of approximately .064 and .052 intercept units per additional year of age, respectively. These positive coefficients indicate that, at the same level of Autonomy, the expected item responses increased with age, conditional on the other moderators. Nordic membership was associated with positive intercept differences for the Pleasure items P3 ($\hat{\beta}_{Nordic} = 0.587$) and P2 ($\hat{\beta}_{Nordic} = 0.487$), indicating higher intercepts relative to Western Europe. Gender-related intercept-moderation coefficients ranged from -0.180 to 0.020, indicating variation in item intercepts between women and men after accounting for the latent factors and other moderators.

SEM trees

Invariance in the CASP-12 measurement model was evaluated separately for loadings and intercepts using score-based instability tests within loading- and intercept-focused SEM trees, following the same logic as in the simulation.

The baseline four-factor model used for the SEM tree analyses converged successfully (N = 65,214, 42 free parameters, (-2LL) = 1,801,369, AIC = 236,317, BIC = -6,873,268). The score-guided SEM tree analyses also indicated parameter heterogeneity at both the metric and scalar levels ($p_{metric} = < .001$ and $p_{scalar} = < .001$, respectively; see see 5 &6).

For metric invariance, cultural cluster was selected as the first partitioning variable by the trees, WesternandNordic separating Western and Nordic countries from Southern and Eastern countries (LR = 5,989.44; df = 8). Within both resulting branches, the next partition was based on age, at approximately 68.5 years. Further age-based partitions occurred at approximately 65 and 79 years, together with an additional distinction between Western and Nordic countries within one subgroup.

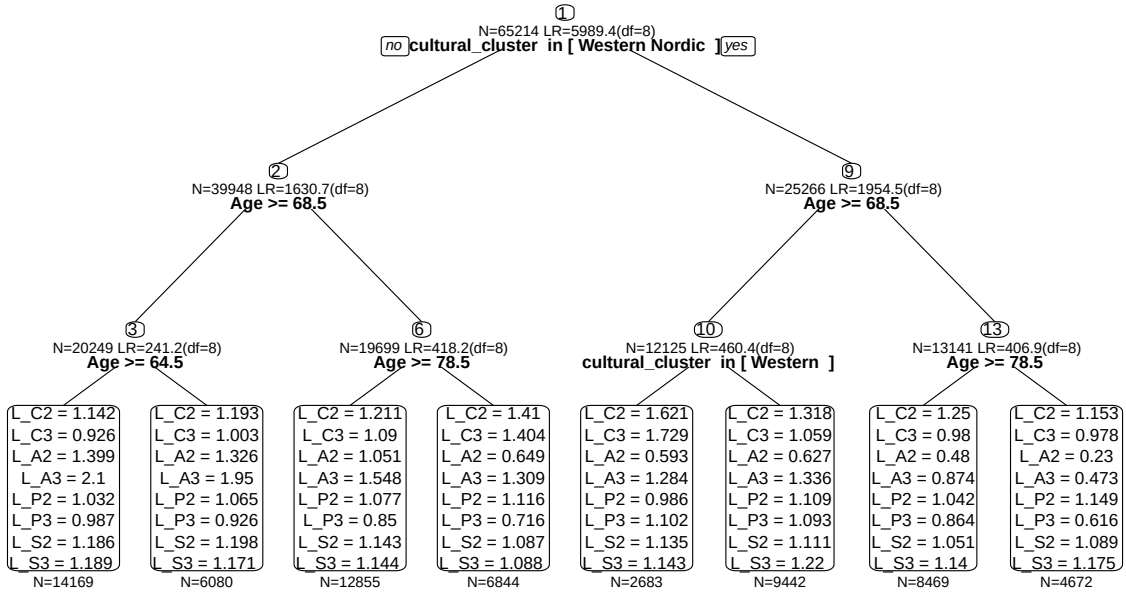


Figure 5

Note. The tree identifies subgroup differences in CASP-12 factor loadings. Internal nodes show splitting variables, likelihood-ratio (LR) statistics, and sample sizes (N); terminal nodes report estimated loadings for the eight non-anchor indicators.

Thus, the metric tree indicated that loading differences were primarily structured by cultural cluster and age. Gender was not selected as a partitioning variable.

For scalar invariance, age was selected as the first partitioning variable, with the initial split occurring at approximately 78.5 years. Cultural cluster was subsequently selected in both age-defined branches, consistently distinguishing Southern Europe from the remaining cultural clusters. Further age-based partitions occurred at 68.5 and 69.5 years among younger participants and at 86.5 and 87.5 years among older participants. Gender was not selected as a partitioning variable.

The scalar tree therefore indicated a combination of age- and culture-related intercept

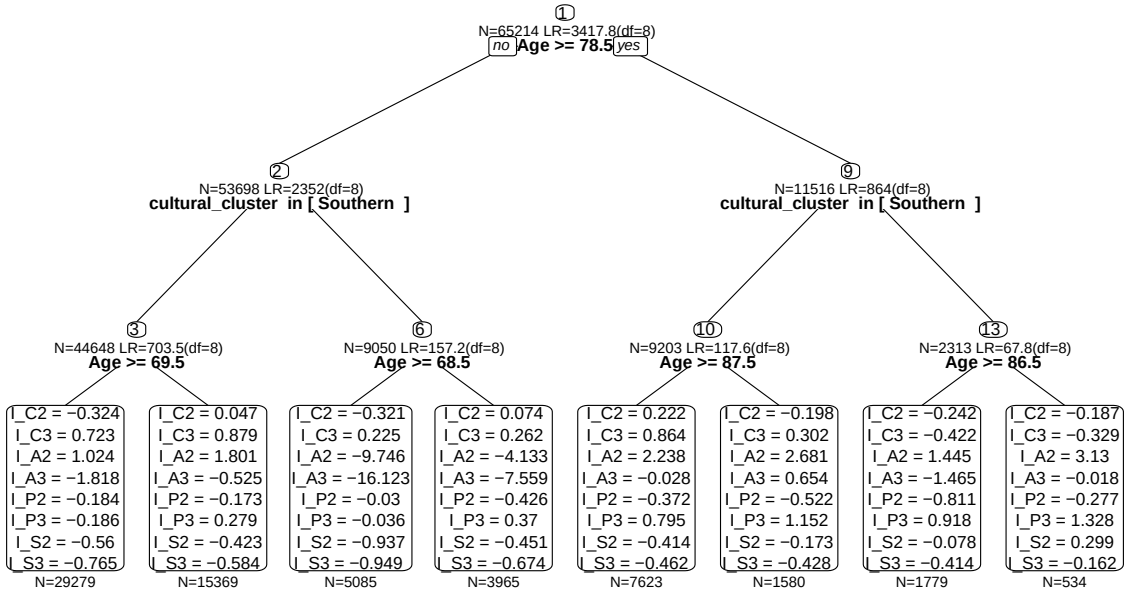


Figure 6

Note. The tree identifies subgroup differences in CASP-12 item intercepts. Internal nodes show splitting variables, likelihood-ratio (LR) statistics, and sample sizes (N); terminal nodes report estimated intercepts for the eight non-anchor indicators.

instability, with Southern Europe emerging repeatedly as a distinct subgroup.

Partial Dependence Plots

The PDP show the factor loadings, predicted by the SEM forest, across age while averaging over the remaining predictors. The PDPs were generated based on 10 trees with a depth of 3 and a case-minimum of 500.

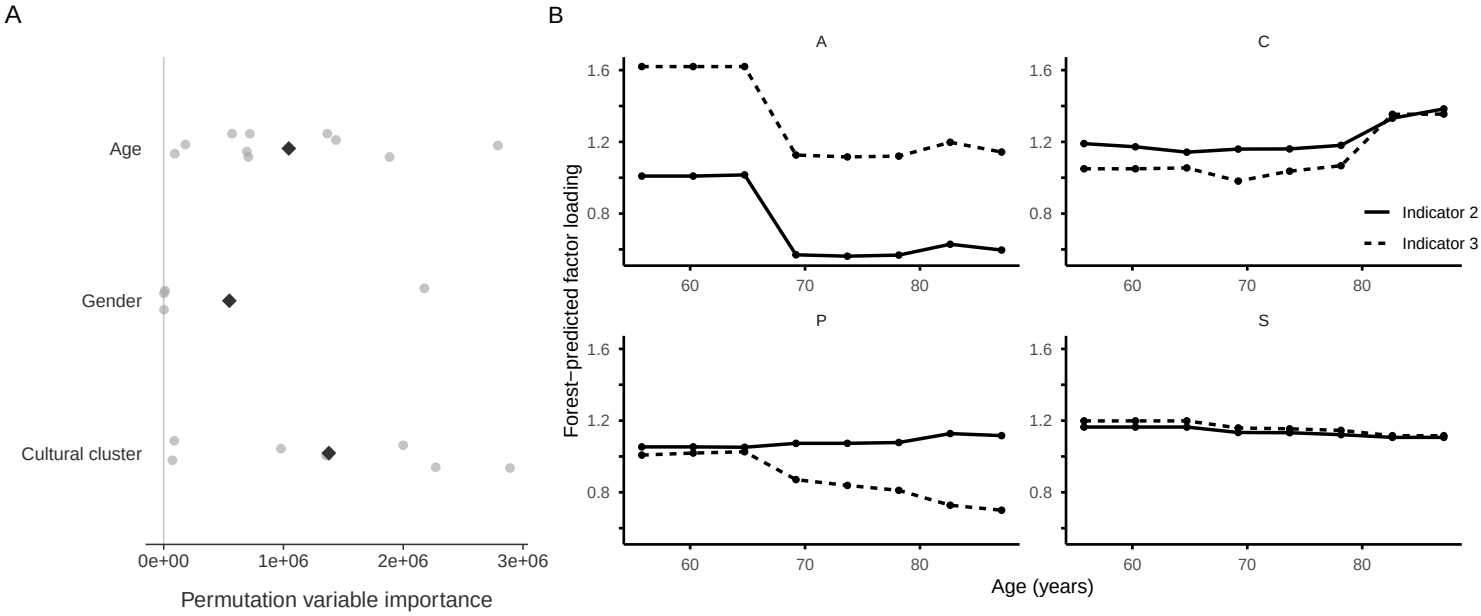


Figure 7

Note. Panel A shows permutation variable importance for age, gender, and cultural cluster, with gray points representing estimates from individual trees and black diamonds indicating mean importance across the 10-tree SEM forest. Panel B shows partial dependence plots of forest-predicted factor loadings for the non-anchor indicators (2 and 3) of Control (C), Autonomy (A), Pleasure (P), and Self-Realization (S). Solid and dashed lines represent indicators 2 and 3, respectively. Age is displayed in years, restricted to the 2.5th–97.5th percentiles of the observed age distribution.

The PDPs suggest that some of the inspected loadings might not be modearted linearly by age. For the items A2 and A3, loadings were rather stable at the lower age range and decreased with age, the largest difference occuring between mid-60s and early 70s. For the factor Pleasure the pattern seems to be more heterogeneous, as the loading of item P2 seems relatively stable across Age, whereas item P3 shows a more pronounced decline, also starting in mid 60s.

Discussion

The present study compared MNLFA and SEM trees for detecting MNI when moderator effects differed in their functional form, magnitude and with different sample sizes. The simulation results revealed a trade-off between model-based specificity and data-driven flexibility. MNLFA achieved high power when the specified moderation function corresponded to the underlying functional form of MNI, but its performance deteriorated substantially under functional-form misspecification. This finding is consistent with previous methodological work highlighting the sensitivity of parametric approaches to assumptions about the functional form of moderation. In particular, Bauer (2023) emphasized that correctly specified parametric functions may improve accuracy and statistical power, whereas incorrect specifications can introduce bias. Similarly, recent work on related methods has highlighted the risk of misrepresenting complex nonlinear moderation patterns when the assumed functional form does not adequately capture the underlying relationship (Liu et al. 2025). The present findings extend these considerations by demonstrating substantial reductions in MNI detection power under functional-form misspecification, especially for MNLFA. SEM trees, in contrast, showed comparatively stable power across linear, quadratic, and sigmoid moderator functions, but were less specific in distinguishing the level at which MNI occurred. Thus, the methods ability to detect measurement heterogeneity, depends on whether its functional form is well assumed in advance and whether localizing MNI to loadings or intercepts was of primary interest. For MNLFA, the dependence on functional form specification was particularly pronounced. Linear MNLFA performed well for linear and sigmoid effects but poorly for quadratic effects, whereas the quadratic specification showed the complementary pattern. The strong performance of linear MNLFA under sigmoid moderation should, however, be interpreted in light of the specific moderator range and steepness used in the data-generating model. Within the simulated range, a substantial portion of the sigmoid relationship was approximately linear, allowing the linear specification to capture much of the induced moderation. Its performance could thus depend on the

extent to which nonlinearity is expressed over the observed moderator range. Additionally, it is likely that additional statistical information cannot compensate for a moderator function that inadequately represents the underlying relationship. As SEM trees, showed consistently high power across linear, quadratic, and sigmoid forms, their performance seems to depend less strongly on correctly anticipating the functional form of MNI. This advantage could be particularly relevant in applications in which theory identifies plausible sources of measurement heterogeneity but provides limited guidance regarding the shape of their association with measurement parameters.

The empirical application complemented these simulation findings. Both MNLFA and SEM trees provided evidence against metric and scalar MI of the CASP-12 in the ninth waves SHARE data and pointed to age and cultural background as relevant sources of measurement heterogeneity, whereas gender-related differences were comparatively limited. The two approaches differed, however, in how this heterogeneity was represented. Where MNLFA quantified moderator-specific changes in individual loadings and intercepts, SEM trees instead identified subgroups defined by combinations of cultural cluster and data-driven age cutoffs. Cultural cluster provided the first partition for loading instability, whereas age provided the first partition for intercept instability, with subsequent splits revealing conditional combinations of age and cultural background. The empirical findings should also be considered in relation to Bonneville-Roussy et al. (2024), whose analysis of the CASP-12 provided the substantive motivation for the present application. Their results for earlier waves of the SHARE data supported the four-factor structure of the CASP-12 and indicated full gender invariance, largely age-invariant measurement, and partial scalar invariance across European cultural groups. Our results also support the general model structure, exhibiting satisfactory model fit, with generally high factor loadings. Regarding the MI analyses, both methods provided little evidence of gender-related measurement heterogeneity, aligning with the findings from Bonneville-Roussy et al. (2024). Second, both pointed to cultural differences in the measurement of the CASP-12, suggesting that cultural background

represents a more relevant source of MNI than gender.

To explore potential functional forms of moderation in factor loadings, using SEM tree based approaches, we estimated a SEM forest with age, gender, and cultural cluster as candidate moderators. The forest was specified to focus on the eight freely estimated factor loadings of the four-factor measurement model. Permutation variable importance was used to explore the relative predictive relevance of the candidate moderators. The PDPs were subsequently calculated for age as an example case to examine potential nonlinear patterns in the model-predicted factor loadings. In particular, the limited number of trees may affect the stability of the estimated relationships, while the tree-based estimation procedure can produce discontinuities that do not necessarily reflect substantive thresholds. Furthermore, PDPs average model predictions across the distribution of the remaining predictors and may therefore involve poorly supported predictor combinations when moderators are correlated. Nonetheless, the PDPs provide some exploratory insights into potential age-related differences in the measurement of CASP-12. Most notably, the predicted loadings of both Pleasure indicators decreased with age, with the most pronounced changes occurring between the mid-60s and early 70s. This pattern may suggest that the aspects of pleasure captured by these indicators become less strongly associated with the underlying construct at older ages. For Self-Realization, the predicted loadings were comparatively stable across most of the age range, although both indicators showed an increase at more advanced ages. This could indicate that the relationship between these indicators and the underlying dimension of self-realization becomes somewhat stronger in later life. The patterns for Autonomy were more heterogeneous, while the loading of one indicator remained relatively stable, the other decreased considerably with age. This may suggest that particular aspects of autonomy change in their relevance for representing the latent construct, rather than the dimension as a whole being affected uniformly. In contrast, the loadings of both Control indicators remained largely stable across age, suggesting comparatively consistent measurement relationships for this dimension. Taken together, these exploratory findings

indicate that potential age-related measurement non-invariance may be concentrated in particular dimensions and indicators, rather than reflecting a general change in the measurement structure of quality of life across age.

Bonneville-Roussy et al. (2024) examined age invariance across four predefined age groups: younger than 60, 60–69, 70–79, and 80 years and older. Whereas they found cross-sectional age invariance for 11 of the 12 CASP-12 items, both MNLFA and SEM trees identified age-related parameter heterogeneity in the present data. MNLFA modeled age continuously, while SEM trees identified data-driven thresholds at which heterogeneity was most pronounced. Several thresholds aligned with the predefined age-group boundaries: the metric tree identified splits around 65, 69, and 79 years, and the scalar tree around 69, 79, and 87–88 years. However, the additional splits around 65 and 87–88 years suggest heterogeneity within the predefined 60–69 and 80-and-older groups, respectively, whereas the boundary at 60 years was not reproduced. MNLFA identified significant age-related differences in both factor loadings and intercepts, with particularly pronounced coefficients for A2 (“Wie oft haben Sie das Gefühl, keinen Einfluss darauf zu haben, was mit Ihnen geschieht?”) and A3 (“Wie oft denken Sie, dass fehlendes Geld Sie davon abhält, die Dinge zu tun, die Sie tun möchten?”). The findings also showed some item-level convergence with Bonneville-Roussy et al. (2024), who identified age-related MNI in an Autonomy item (“Do the things you want to do.”; A1). Similarly, the present MNLFA results showed comparatively pronounced age-related moderation for Autonomy items A2 and A3 (item A1 was used for identification). Although this suggests that Autonomy warrants particular attention, age-related heterogeneity was not confined to this dimension.

Methodological Discussion

MNLFA is inherently model-based, researchers specify which covariates are expected to moderate measurement parameters and how these moderator effects enter the model. When these assumptions are well founded, the resulting coefficients provide a direct and

parsimonious description of how individual measurement parameters vary with the covariates. SEM trees shift part of this specification from the researcher to the data. Rather than requiring a particular functional form or predefined cutoffs, they search for partitions of the covariate space associated with parameter heterogeneity. Consequently, the resulting tree describes where in the covariate space the measurement model changes, including potentially conditional combinations of moderators, rather than estimating a single prespecified moderation function. These approaches should therefore not be viewed simply as interchangeable tests competing on statistical power. They provide complementary ways of characterizing MNI, and their usefulness depends on the substantive question and the amount of prior knowledge available (Goretzko et al. 2026).

This distinction between detection and localization is important for interpreting the comparative performance of the methods. The high overall power of SEM trees indicates that they are effective at identifying heterogeneity in the measurement model across the conditions considered, but a detected instability should not necessarily be interpreted as uniquely identifying the parameter class responsible for that heterogeneity. Notably, the SEM-tree metric test frequently rejected invariance when only intercepts were noninvariant, whereas scalar rejection generally remained near the nominal level when only loadings were noninvariant. This asymmetric cross-level sensitivity was particularly pronounced at larger sample sizes in some scalar-only conditions, suggesting systematic sensitivity to intercept instability rather than sampling imprecision.

This cross-level sensitivity also affected the sequential classification of MNI. SEM trees correctly classified metric-only MNI in 80.9% of replications and combined metric and scalar MNI in 77.4%, but scalar-only MNI in 69.6%, compared with 63.6% for linear MNLFA and 32.9% for quadratic MNLFA. Under scalar-only MNI, erroneous rejection at the preceding metric stage occurred in 25.8% of SEM-tree replications, 11.1% for linear MNLFA, and 26.5% for quadratic MNLFA, preventing these cases from reaching scalar testing. As

illustrated in Figure 4, this effect varied substantially across moderator forms and effect magnitudes. However, premature metric rejection did not fully explain the differences in classification accuracy, particularly for linear MNLFA, indicating that errors at the subsequent scalar stage also contributed. Thus, sequential classification rates reflect not only sensitivity to MNI but also the ability to distinguish loading from intercept instability.

One way to deal with this in general could be to test all levels of invariance, that are of relevance to ones research question, regardeless of rejection on early levels.

Relation to CFA Trees

The SEM-tree analyses used in the present study are closely related to CFA trees (Goretzko et al. 2026) and represent a particular specification of the broader SEM-tree framework rather than a distinct method. Both embed a prespecified confirmatory factor model within model-based recursive partitioning. In the present study, metric and scalar MNI were investigated using score-based instability tests focused on factor loadings and item intercepts, respectively, while other free parameters were not globally fixed across the tree. By contrast, the intercept-focused CFA-tree approach described by Goretzko et al. (2026) fix nonfocal parameters to estimates from a single-group CFA, thereby more explicitly isolating intercept instability tests.

This distinction is relevant to the present findings. Despite focusing the metric instability test on loadings, intercept-only MNI produced elevated metric rejection rates in several conditions. Thus, restricting the focal parameters did not ensure that the test responded exclusively to loading heterogeneity. More strongly constraining nonfocal parameters, as in the CFA-tree approach, might improve parameter-level attribution, although the present simulation did not examine whether this would reduce cross-level sensitivity or introduce other forms of misspecification.

A potential direction for future research could be to investigate a sequential MI

procedure in which parameter constraints are progressively relaxed. Rather than beginning with a loading-focused SEM tree in which nonfocal parameters remain freely estimated, metric invariance could initially be evaluated using a more strongly constrained CFA-tree specification, as described by Goretzko et al. (2026). By fixing nonfocal parameters, this approach would aim to isolate loading instability and reduce the possibility that heterogeneity in other model parameters influences the metric test. If metric invariance is retained, subsequent scalar testing could relax selected constraints while evaluating instability in item intercepts. Such a strategy would reverse the popular constraint structure examined in the present study, moving from a more restricted metric analysis toward a more flexible scalar analysis. Whether stronger constraints improve the localization of MNI however remains to be seen. The present findings demonstrate that cross-level sensitivity can arise even when nonfocal parameters remain freely estimated, indicating that parameter-focused instability tests do not necessarily isolate heterogeneity in the corresponding parameter class. Conversely, fixing nonfocal parameters may introduce misspecification when those parameters are genuinely noninvariant, potentially producing similar cross-level effects. Thus, neither strategy guarantees that detected instability can be attributed exclusively to the focal parameters.

An additional consideration is the identification of latent means. Because observed subgroup differences can arise from latent mean differences, intercept noninvariance, or both, their attribution depends on the constraints imposed on the model. As discussed by Goretzko et al. (2026), fixing latent means may cause genuine differences in the latent construct to appear as intercept instability, whereas constraining intercepts while allowing latent means to vary may produce the opposite attribution. Consequently, stronger parameter constraints cannot automatically be assumed to yield more accurate identification of MNI. Future research should examine how alternative constraint and identification strategies affect both cross-level sensitivity and the distinction between measurement noninvariance and substantive latent mean differences.

These findings also support the integration of tree-based procedures into broader MI assessment workflows rather than their use as standalone methods. Tree-based approaches are particularly useful when potential moderators, nonlinearities, interactions, or cutoffs are not known in advance, whereas MNLFA is advantageous when moderator effects can be theoretically specified and parameter-level estimates are of substantive interest. Rather than viewing these methods as competing alternatives, Goretzko et al. (2026) recommends integrating tree-based procedures into established MI assessment workflows, such as those discussed by Somaraju et al. (2022), alongside approaches such as MGCFA or MNLFA. Trees can identify previously unspecified sources and patterns of heterogeneity, which can subsequently be examined using explicitly parameterized measurement models. Such complementary analyses are particularly valuable given the cross-level sensitivity observed in the present study, which cautions against interpreting tree-based instability as unambiguous evidence of a specific level of MNI. Furthermore the exploratory PDPs in this study suggest that they could also complement a tree based approach to MI-testing by approximating the functional form of the moderation. However, one should try to grow a stable forest, as to not overestimate results from a computationally constrained SEM forest, like in this study. The resulting patterns should then be interpreted as preliminary indications of moderation patterns rather than definitive estimates of the underlying functional forms.

Regularization offers a further extension to the presented approaches and can be particularly useful for complex measurement models or when many potential sources of MNI are under consideration. Within exploratory factor analyses, regularization can be used to obtain sparse and interpretable loading structures without relying on conventional factor rotation (Goretzko 2025), whereas within MNLFA, regularization has been applied to moderation parameters to identify measurement bias across multiple categorical and continuous background characteristics (Bauer et al. 2020; Belzak and Bauer 2024). Although such extensions can reduce the need to prespecify individual sources of measurement heterogeneity, they also introduce additional methodological choices concerning penalization

and model selection. Method choice and combination should therefore be guided by the analytical objective.

Limitations and Future Research

Also please note, that we inspected the ninth wave of SHARE, Whereas Bonneville-Roussy et al. (2024) investigated the waves two and seven. The empirical example therefore illustrates that MNLFA and SEM trees can yield broadly convergent substantive conclusions while providing different descriptions of the underlying measurement heterogeneity.

Several limitations qualify the generalizability of the present findings. First, the simulation considered linear, quadratic, and sigmoidal moderation, representing distinct but not exhaustive forms of MNI. Although SEM trees performed comparatively consistently across these conditions, this advantage cannot be generalized to arbitrary functional relationships. Second, the simulation employed a relatively simple one-factor model with four continuous indicators. Future studies should examine whether the observed performance patterns extend to more complex measurement structures, including multifactor models, ordinal indicators, cross-loadings, and correlated residuals.

The simulation also largely separated the detection of MNI from moderator selection by supplying SEM trees with the relevant moderator variables. Although EFA trees have previously been evaluated in the presence of irrelevant candidate covariates (Sterner and Goretzko (2023)), and methodological work on SEM trees has examined selection bias and multiple-testing adjustments (Arnold et al. (2021)), the additional noise-predictor analysis in the present study addressed these issues only under restricted conditions. Future research should therefore investigate how parameter-focused CFA-based trees perform with larger sets of correlated and irrelevant moderators, particularly regarding the probability of selecting the true source of MNI, the stability of selected variables and split points, and the trade-off between false-positive selections and statistical power. Such investigations are especially

relevant because a selected moderator may act as a proxy for another correlated source of measurement heterogeneity.

The empirical application was intended to illustrate the methods rather than establish definitive age or cultural thresholds in the CASP-12. Importantly, the present study analyzed SHARE wave 9, whereas Bonneville-Roussy et al. (2024) investigated waves 2 and 7. Differences between the findings may therefore reflect variation in sample composition or measurement properties across waves, as well as methodological differences. Although MNLFA and SEM trees yielded broadly convergent substantive conclusions in the present application, they characterized the underlying heterogeneity differently. Replication across SHARE waves and independent samples would help establish the stability of the identified moderators, age partitions, and item-level parameter differences.

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